

# ISLAMIC UNIVERSITY OF TECHNOLOGY

**EEE 4710 – Artificial Intelligence and Machine Learning Lab**

**Group Name: Bangi**

**Topic:** Bangladesh Electricity Consumption Analysis


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| Field | Details |
|---|---|
| Project Title | Bangladesh Electricity Consumption Analysis |
| Course Code | EEE 4710 – AI and Machine Learning Lab |
| Dataset | BPDB Daily Electricity Data – Version 1 (2019–2024) |
| Problem Type | Supervised Regression |
| Target Variable | Evening-Peak Electricity Demand (MW) |
| Algorithms Used | Linear Regression, Random Forest, Gradient Boosting |
| Source | Bangladesh Power Development Board (BPDB) |

## 1.Problem Definition:

Bangladesh has been going through a prolonged energy transition over the past decade. On one hand, installed generation capacity has expanded substantially — from around 9,000 MW in 2019 to over 16,000 MW by 2024. On the other hand, actual delivered electricity has not kept pace, largely because fuel supply constraints gas shortages, coal import bottlenecks, and declining hydroelectric output have repeatedly kept a large share of that capacity offline. The result is a chronic mismatch between what the grid can theoretically supply and what consumers actually receive, and load shedding has remained a persistent problem, especially during summer months when cooling demand spikes.

The core question this project tries to answer is straightforward: given what we know about a day the temperature, the availability of fuel, the number of plants that are running, and the time of year can we predict how much electricity the national grid will need to deliver at the evening peak? Evening-peak demand is particularly important because it is the daily maximum that grid operators must plan against; getting that number wrong means either overcommitting generation reserves unnecessarily or failing to prevent a blackout.

This is a well-defined supervised regression problem. We have five years of daily historical records from the Bangladesh Power Development Board (BPDB), each containing the inputs described above alongside the observed peak demand. Machine learning models — especially tree-based ensemble methods — are well suited to this kind of structured tabular data, and the dataset is large enough to train and validate them properly.

## 2.Dataset Preparation:

The data used in this project is Version 1 of the structured daily electricity dataset .There is a single Excel file with 1,848 rows (one per calendar day) and 41 columns covering the period from 21 November 2019 to 30 December 2024.

Each row records national-level statistics peak demand, highest generation, minimum generation, day-peak and evening-peak generation, a morning generation forecast, maximum temperature in Dhaka, and three supply-constraint variables (gas/liquid fuel limitation, coal supply limitation, and low water level at Kaptai hydroelectric plant). It also records the number of generation plants that were offline for maintenance or shutdown on that day. Additionally, the dataset includes zone-wise demand, supply, and load shedding figures for nine administrative divisions: Dhaka, Chattogram, Rajshahi, Khulna, Rangpur, Mymensingh, Sylhet, Barishal, and Cumilla.

| Attribute | Details |
|---|---|
| Time Coverage | 21 November 2019 – 30 December 2024 |
| Total Records | 1,848 daily entries |
| Total Features | 41 columns |
| Target Variable | Max. Demand at eve. peak – Generation end (MW) |
| Geographical Scope | National level + 9 administrative divisions |
| Missing Data | Khulna region: 61 missing entries; Temperature: 5; others: 1–9 |
| Data Version Used | Version 1 – Raw Extract (unprocessed, direct PDF parse) |
| Dataset DOI | 10.17632/x7r7wdb39k.1 (Mendeley Data) |

## 2.1 Data preprocessing:

### 2.1.1 Date Parsing and Temporal Features

The date column was stored as a string in day.month.year format .It was parsed into proper datetime objects and the dataset was sorted chronologically. Four temporal features were then derived: Year, Month (1–12), Day of Week (0–6, where 0 is Monday), and a binary IsWeekend flag. In Bangladesh, the working week runs Sunday through Thursday, so Friday and Saturday were coded as weekend days (IsWeekend = 1).

### 2.1.2 Handling Missing and Invalid Values

Zero entries in the peak demand column are physically impossible  if demand were truly zero, there would be no electricity grid to speak of. These were replaced with NaN and treated as missing.All numerical missing values were filled using forward-fill (carrying the previous valid reading forward) followed by backward-fill for any remaining gaps at the beginning of the series. This is the simplest reasonable approach given the daily-record nature of the data.

### 2.1.3 Outlier Removal

Outliers in the demand column were removed using the standard IQR method. Any demand value more than 1.5 interquartile ranges below the first quartile or above the third quartile was dropped. The final clean dataset used for analysis contained 1,844 rows.

### 2.1.4 Feature Scaling

Tree-based models (Random Forest and Gradient Boosting) are insensitive to the scale of input features, so no transformation was applied for them. For the Linear Regression model, all features were standardised to zero mean and unit variance using scikit-learn's StandardScaler to ensure that coefficients were comparable and gradient descent-based solvers converged properly.

## 2.2 Exploratory Data Analysis:

### 2.2.1 Long-Term Demand Growth

Average evening-peak demand grew steadily every year, from roughly 8,496 MW in 2019 to approximately 12,950 MW in 2024 — an increase of about 52% over five years. The year 2019 is only a partial year (starting November), so its average is pulled somewhat lower than it would otherwise be. The growth trend is broadly linear across full calendar years (2020–2024), reflecting Bangladesh's rapid economic expansion and rising electricity access rates during this period.

### 2.2.2 Seasonal Variation

Demand shows a clear and consistent seasonal cycle. The lowest monthly average is in December (around 9,303 MW), when temperatures are cool and air-conditioning load is minimal. Demand then climbs through the spring, peaks in September (around 13,468 MW), and falls back in the autumn. It might seem counterintuitive that September, rather than April or May, is the peak month — but this likely reflects the combination of still-high temperatures and the delayed effect of monsoon-related agricultural and industrial activity in the second half of the year. The months of July, August, and September consistently account for the highest demand figures.

### 2.2.3 Temperature and Demand

The Pearson correlation between daily maximum temperature in Dhaka and evening-peak demand is 0.68, which is a moderately strong positive relationship. A scatter plot coloured by year makes the structure clearer: the upward drift in demand across years is visible as distinct clusters, but within any given year the temperature-demand relationship holds. Higher temperatures drive up residential and commercial cooling load, which translates into higher grid demand. The relationship is not purely linear, and there is substantial scatter, which is why temperature alone is not sufficient to predict demand — but it is an important feature.

### 2.2.4 Load Shedding

Total national load shedding was essentially zero in 2019 and remained low through 2020 and 2021. It then jumped sharply in 2022, when the average daily load shedding across all nine divisions combined reached approximately 234 MW. This closely tracks the global energy market

disruption following Russia's invasion of Ukraine in early 2022, which drove up liquefied natural gas prices sharply and made it uneconomical for Bangladesh to import sufficient fuel to run gas-fired turbines at full capacity. Load shedding remained elevated through 2023 and 2024 as gas limitations worsened further.

### 2.2.5 Regional Breakdown

Dhaka dominates national electricity consumption by a wide margin, accounting for well over double the demand of the next largest division (Chattogram). This reflects Dhaka's role as both the country's capital and its primary industrial and commercial hub. Smaller divisions — Barishal, Mymensingh, and Rangpur — have much lower absolute demand figures. This regional imbalance is important context for any policy analysis: grid investments and load management interventions in Dhaka have a disproportionate impact on national statistics.

### 2.2.6 Day-of-Week Effect

Average demand on Fridays and Saturdays (the Bangladeshi weekend) is noticeably lower than on weekdays. This is consistent with reduced commercial and industrial activity during the weekend. The difference, while not dramatic, is statistically meaningful and justifies including the day-of-week and IsWeekend features in the model.

### 2.2.7 Generation–Demand Gap

By comparing maximum daily generation against the recorded demand, we can trace the surplus or deficit at the national level over time. In 2019 and early 2020, the grid generally ran a small surplus generation exceeded demand on most days, meaning there was headroom. From 2022 onwards, periods of deficit became more frequent, corresponding exactly to the years of severe fuel shortages. The gap analysis makes the energy crisis of 2022–2024 visually clear: it was not primarily a problem of insufficient installed capacity, but of fuel availability preventing that capacity from being dispatched.

### 2.2.8 Fuel Constraint Trends

Gas and liquid fuel limitations have worsened substantially over the study period. In 2019, the average daily gas limitation was around 1,233 MW — meaning that 1,233 MW of potential

generation was unavailable due to inadequate fuel supply. By 2024 this had risen to approximately 3,301 MW per day. Overlaying gas limitation against demand on the same chart makes a striking point: the two are not strongly correlated with each other. Demand kept growing regardless of fuel constraints; it was the load shedding that absorbed the difference.

#### 2.2.9 Smoothed Demand Trend

A 30-day centered rolling average was applied to the daily demand series to strip out short-term noise and holiday effects and reveal the underlying trend more clearly. The smoothed series confirms the steady year-on-year growth, the consistent summer peaks, and the winter troughs. It also shows that there was a brief but noticeable dip in demand in early 2020 that corresponds to the COVID-19 lockdown period, when industrial and commercial electricity consumption dropped sharply for several months.

## 3. Model development:

Three regression algorithms were trained and compared. The choice was intentional: Linear Regression serves as the simplest baseline; Random Forest provides a robust non-linear ensemble that handles feature interactions without much tuning; and Gradient Boosting is typically the strongest performer on tabular data and acts as the upper bound for what an off-the-shelf model can achieve on this dataset.

| Model | Type | Key Settings |
|---|---|---|
| Linear Regression | Parametric, linear | Features standardised with StandardScaler |
| Random Forest | Ensemble, bagging | 100 decision trees, random_state = 42 |
| Gradient Boosting | Ensemble, boosting | 100 boosting stages, random_state = 42 |

### 3.1 Feature Selection

Ten features were selected as model inputs. The meteorological input is daily maximum temperature in Dhaka. Supply-side constraints are captured by three variables: gas/liquid fuel limitation (MW), coal supply limitation (MW), and low water level at Kaptai lake (MW).

Operational status is captured by the number of plants offline. Generation capacity is represented by the highest generation recorded that day and the morning generation forecast. Finally, three temporal features were added: month (1–12), day of week (0–6), and a binary weekend flag. These features were chosen because they are all available in practice before the evening peak occurs and because each has a plausible causal relationship with demand.

### 3.2 Train-Test Split

The dataset was split 80% training and 20% testing using a random split with a fixed seed (random_state = 42) to ensure reproducibility. All 1,844 clean records were used; no time-based split was applied at this stage. Additionally, 5-fold cross-validation was performed on the best-performing model to verify that results were not an artefact of a lucky random split.

## 4. Performance evaluation:

All three models were evaluated on the held-out test set using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ($R^2$). MAE gives the average absolute prediction error in the same units as demand (megawatts), which is easy to interpret directly. RMSE penalises large errors more heavily than MAE, making it sensitive to outlier predictions. $R^2$ measures what fraction of the total variance in demand the model explains.

| **Model** | **MAE (MW)** | **RMSE (MW)** | **$R^2$ Score** |
|---|---|---|---|
| Linear Regression | 219.2 | 455.4 | 0.9518 |
| Random Forest | 195.6 | 458.1 | 0.9512 |
| Gradient Boosting | 206.7 | 430.9 | 0.9569 |

All three models performed similarly and very well, with $R^2$ scores above 0.95. This is a high result, but it is not surprising given the structure of the data. Maximum generation (Max_Gen) alone contributes roughly 96% of the feature importance in the Random Forest model, which reflects an important reality about Bangladesh's electricity sector: because the grid is supply-constrained,

the demand that actually gets delivered is largely determined by how much the plants can generate on a given day. In other words, the model is partially learning the identity function demand tends to equal supply in a constrained grid.

Gradient Boosting had the best RMSE (430.9 MW), followed closely by Linear Regression (455.4 MW) and Random Forest (458.1 MW). The fact that Linear Regression performs almost as well as the ensemble models further confirms that once generation capacity is included, the remaining variation in demand is nearly linearly predictable. The ensemble methods have slightly lower MAE values, suggesting they handle individual peak days better, but the difference is not substantial.

### 4.1 Cross-Validation

Five-fold cross-validation on the Gradient Boosting model produced $R^2$ scores of [0.964, 0.972, 0.894, 0.937, 0.949] across the five folds, with a mean of 0.943 and a standard deviation of 0.028. The third fold had a somewhat lower score than the others, which is likely explained by the fact that a random split placed a disproportionate number of anomalous 2022 energy-crisis days in the test portion of that fold. Overall the cross-validation confirms that the model generalizes well and is not overfitting to a particular split.

### 4.2 Residual Analysis

The residuals from the Gradient Boosting model are roughly centred around zero and approximately normally distributed, which is a good sign. There is no strong systematic pattern in a plot of residuals against predicted values, meaning the model is not systematically underestimating or overestimating demand across the prediction range. The largest residuals tend to occur at very high demand values (above 15,000 MW), which correspond to the unusual peak days in 2023 and 2024 when demand reached its historical maximum. Those days are harder to predict because they represent conditions outside the historical norm.

### 4.3 Feature Importance

The Random Forest feature importance scores make the dominance of generation capacity very clear: Max_Gen accounts for approximately 0.956 of the total importance score, and Gen_Forecast contributes another 0.030. All other features — temperature, gas limitation, coal

limitation, water level, plants offline, month, day of week, and weekend flag — collectively account for only about 1.4% of importance. This does not mean those features are useless; they likely contribute to the model's ability to predict demand on days when generation capacity is ambiguous. But it does confirm that the single most important predictor of how much electricity Bangladesh will consume on a given evening is how much the power plants can physically produce.

## 5. Discussion:

The results raise an interesting question: if generation capacity is such an overwhelming predictor of demand, is this really a demand forecasting model, or is it closer to a generation dispatch model? The answer is that in a supply-constrained power system, these two things are not as distinct as they would be in a well-functioning grid. In a country like Bangladesh where load shedding is common, measured 'demand' is actually met demand the electricity consumers actually received rather than the electricity consumers wanted. The unmet portion disappears into load shedding statistics and never shows up in the demand column. A model that learns this dataset is therefore learning a lot about supply, and that is a legitimate and useful thing to model for grid planning purposes.

A second observation worth making is that temperature and fuel availability, despite their low feature importance scores in the Random Forest, are actually important causal drivers of demand. The reason they score low is that once you know the generation capacity, you already have a very tight bound on what demand can be. Temperature and fuel constraints are partly captured within the generation capacity variable itself when it is very hot, generation capacity tends to be dispatched at near maximum; when fuel is short, plants are offline. The causal chain runs through the generation variables. This does not mean temperature is unimportant; it means the dataset structure partially obscures its individual contribution.

The energy crisis of 2022 shows up clearly across multiple analyses — the load shedding spike, the gas limitation increase, and the widening generation-demand gap all coincide. This validates the dataset's quality and the analysis pipeline's ability to identify real events. Future work that

incorporates fuel price data or LNG spot prices from international markets would likely find strong predictive power for these supply-constraint variables.

### 5.1 Limitations

The most significant limitation of this analysis is that Version 1 data was used with simple forward-fill imputation. The missing values in Khulna (61 rows) and elsewhere were treated crudely, and some of the imputed values are almost certainly inaccurate.

A second limitation is the absence of lag features. Electricity demand has strong day-to-day autocorrelation yesterday's demand is a powerful predictor of today's. None of the models in this project used past demand as a feature, which is a significant omission for a time-series forecasting application. This was a deliberate simplification to keep the project accessible, but any production forecasting system would include lagged demand features.

Finally, the 80/20 random split does not respect the temporal structure of the data. A proper time-series evaluation would use the earlier years for training and the later years for testing, to simulate how the model would perform in a real forward-looking deployment.

### 5.2 Possible Improvements

1. Adding lagged demand features (previous day, previous week, same weekday last week) to capture temporal autocorrelation.
2. Switching to a time-series cross-validation strategy (e.g., sklearn's TimeSeriesSplit) to evaluate the model in a more realistic forward-prediction scenario.
3. Exploring LSTM, GRU, or Temporal Convolutional Network architectures, which are designed for sequential data and can model multi-step ahead forecasts.
4. Incorporating divisional demand data as additional targets or auxiliary features to build a multi-output model that forecasts regional demand simultaneously.
5. Adding external features such as industrial output indices, LNG import prices, or holiday dummies to improve the model's ability to predict demand on anomalous days.